higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 3

(6050433)

14 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only BLUE or BLACK ink.
5. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Commercial fertilisers are categorised according to their NPK content.

A 50 kg bag of fertilisers has an NPK composition of 18:24:6.

1.1.1 Indicate the percentage of potassium in the bag. (1)

1.1.2 State TWO functions of nitrogen in plants. (2)

1.1.3 Name TWO forms of inorganic nitrogen fertilisers. (2)

1.1.4 Briefly explain why phosphorus fertilisers should be introduced to soil rather than to the leaves. (2)

1.1.5 How can one identify a lack of potassium in plants? (3)

1.2 Write down the chemical formulae for the following:

1.2.1 Potassium sulphate

1.2.2 Mono-ammonium phosphate

1.2.3 Superphosphate

(3 × 2) (6)

1.3 List FIVE stages in the production of sulphuric acid in sequence. (5)

1.4 Complete and balance the following equation:

$NO_2(g) + H_2O(\ell) \rightarrow$ (4)

[25]

QUESTION 2

2.1 Match a sodium compound from those the list given here to a description given thereafter. Write only the word or term next to the question number (2.1.1–2.1.5) in the ANSWER BOOK.

$NaC\ell$; Na_2SO_4; NaOH; $NaHCO_3$; Na_2CO_3

2.1.1 Also known as washing soda.

2.1.2 One of its first uses was as a laxative.

2.1.3 It is commonly known as caustic soda.

2.1.4 It is used as a baking powder in the food and beverage industry.

2.1.5 The most common natural sodium compound that can also be used to preserve meat.

(5 × 1) (5)

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2.2 South Africa is a deveopling country with an evergrowing demand for cement in the construction industry.

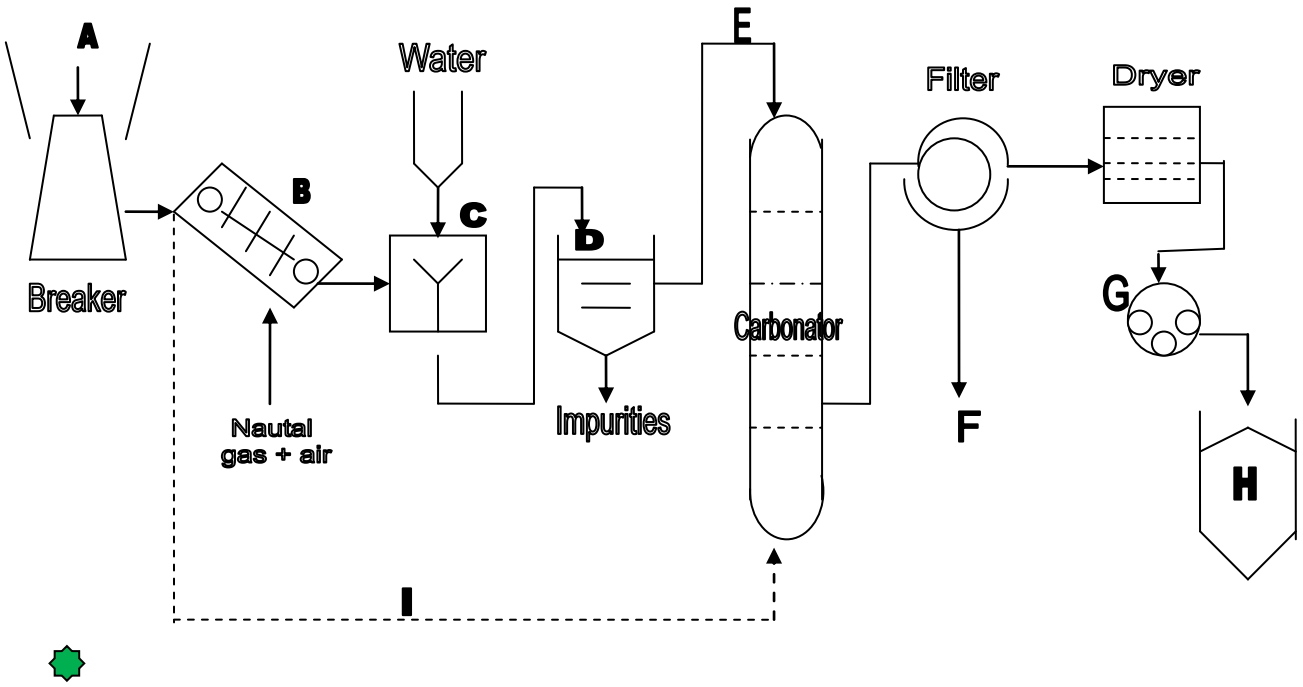
2.2.1 Name TWO types of rock used as raw material in the production of cement. (2)

2.2.2 Explain why the wet process is preferred over the dry process in cement production. (1)

2.2.3 Describe the meaning of *slagment*. (3)

2.2.4 State FIVE advantages of using slagment over normal cement. (5)

2.3 Complete the following process flow diagram of calcium carbonate production by writing the answers next to the letters (A–I) in the ANSWER BOOK.



(9)
[25]

QUESTION 3

Sodium hydroxide (NaOH) does not occur naturally and it is produced through the chlor-alkali process. Produce freehand drawing of the chlor-alkali process. **[10]**

QUESTION 4

4.1 Detergents are more effective than soap, more especially in hard water.

4.1.1 Define the term *hard water*. (2)

4.1.2 Write brief notes on the operation principle of surfactants. (5)

4.1.3 State TWO examples of amphoteric surfactants. (2)

4.2 What are *fillers*? (2)

4.3 Name THREE main raw materials used during the production of soap. (3)

4.4 List SIX steps involved in the production of soap through the continuous process in sequence. (6)

[20]

QUESTION 5

5.1 Name TWO groups of wood used in the production of pulp. (2)

5.2 Write a short description of each of the following areas of a tree trunk.

5.2.1 Bark

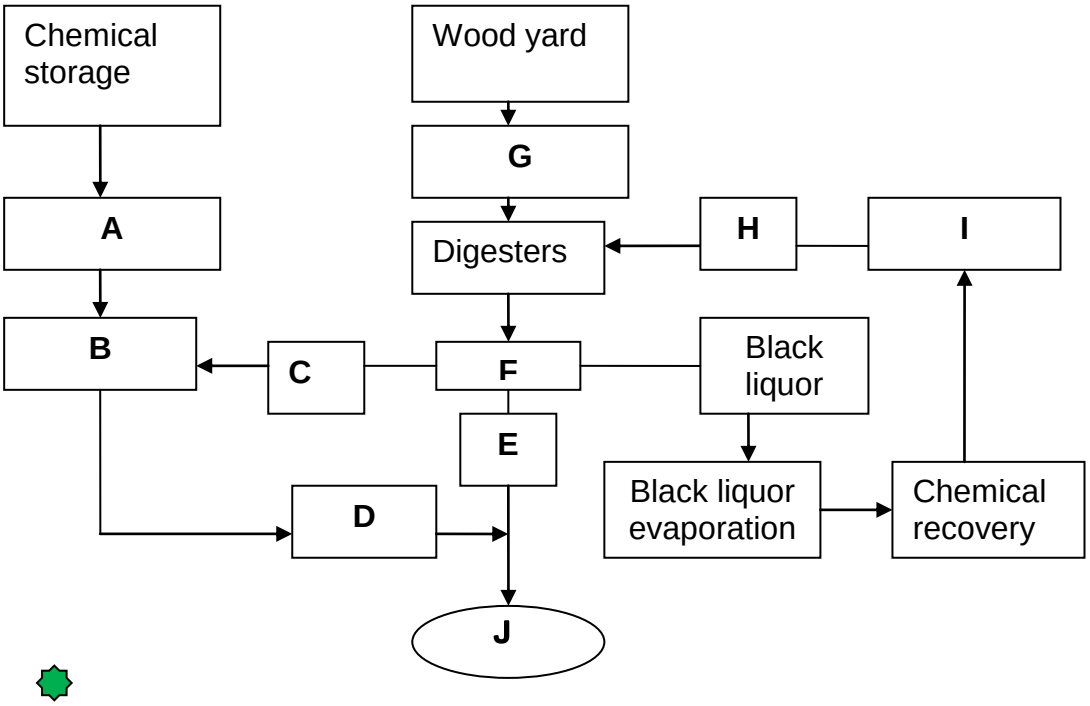
5.2.2 Cambium

5.2.3 Sapwood

5.2.4 Heartwood

(4 × 2) (8)

5.3 Complete the following diagram of a Kraft pulp production process by writing the answer next to the letters (A–J) in the ANSWER BOOK.



(10)
[20]

TOTAL: 100

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